



Ab initio investigation of α - and ζ -V₂O₅ for beyond lithium ion battery cathodes

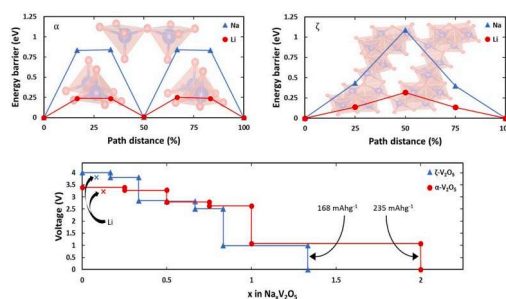
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HIGHLIGHTS

- All ions (Li, Na, Mg, Ca, Al) intercalate into α - & ζ -V₂O₅.
- Van der Waals interactions play larger role in α -V₂O₅ structure evolution.
- Cell voltages vary widely from one ion to another.
- Na has a larger initial voltage than that of Li in both α - & ζ -V₂O₅.
- Na intercalation yields higher specific capacity in α -V₂O₅.

GRAPHICAL ABSTRACT



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ABSTRACT

Mono- (Li/Na), di- (Mg/Ca) and trivalent (Al) ion intercalation is investigated in the α and ζ phases of V₂O₅. Density functional theory is used to calculate the physical and electronic properties of each in the context of secondary battery cathodes. Various exchange-correlation functionals and van der Waals correction schemes are employed in calculating lattice parameters, bandgaps, density of states, cell voltages, and ion diffusion barriers. For intercalations of similar charge transfer (i.e., Li_{0.67}, Na_{0.67}, Mg_{0.33}, Ca_{0.33}, and Al_{0.20}), Ca provides the highest voltage (2.95 V) in α -V₂O₅ with both PBE and PBE-D3. However, Li/Na yields the highest voltage (3.50 V/3.69 V) in ζ -V₂O₅ for PBE/PBE-D3. Monovalent ion diffusion barriers are calculated and show that Li and Na possess barriers of 0.39/0.32 eV and 1.27/1.09 eV in α / ζ -V₂O₅, respectively. A full voltage profile for Na intercalation in each phase determined the initial voltage in ζ -V₂O₅ (4.00 V) is higher than that in α -V₂O₅ (3.4 V), but lacks in specific capacity (168 mAhg⁻¹ versus 235 mAhg⁻¹). With a high diffusion barrier and low specific capacity, ζ -V₂O₅'s utility as a sodium ion battery is bleak. However, both phases show promise for a lithium ion battery application.

1. Introduction

The proposal and development of the Li ion battery (LIB) [1–4] was revolutionary in the field of energy storage. Its unprecedented high energy/power density, gravimetric/volumetric capacity and cell

potential made it the premiere choice for portable electronics. Despite this, our demand for improved battery performance continues to grow concurrently with awareness of the pitfalls of LIBs (i.e., lack of elemental abundance, costs, stability, and environmental impact [5,6]). To circumvent these issues, investigations into viable replacement cathode

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materials for LIBs [7] have been undertaken at the same time as investigations into various ion species, both mono- and multivalent (MV), in hopes of replacing Li altogether [8–10]. Transition metal oxides are commonly investigated as cathode materials due to the multiple available oxidation states of the transition metal and structural framework yielding intercalation channels. Possessing more than one available oxidation state allows the transfer of multiple electrons during intercalation, that is, intercalation of more than one Li ion or various multivalent ion species (e.g., Mg, Ca, and Al). Vanadium is one such transition metal.

Vanadium oxides (VOs) have been shown to possess several properties conducive to their application as secondary battery cathodes. First and foremost, VOs have been shown to possess the ability to operate as electrodes for many of the most rigorously studied metal-ion batteries (e.g., Li, Na) [9,11–16]. Other properties include (but are not limited to) the ability to intercalate MV ions [9], reversible metal-insulator transition [17], and ionic and electrical conductivity [18]. A layered VO (α - V_2O_5) has been extensively studied for use in LIBs and other secondary battery applications [9,10,13–16,19–33]. However, limitations such as gravimetric capacity, insertion energy, and lattice distortions (to name a few) appear to hinder the advancement of α - V_2O_5 for LIBs and many propose utilizing other ion species and/or VO polymorphs [9–16, 19–35]. One such polymorph is a tunneled-structure, ζ - V_2O_5 . Unlike the α phase, ζ - V_2O_5 is far less studied, especially in a computational context. To the best of the authors' knowledge, only one other computational work [14] has reported lattice parameters for ζ - V_2O_5 and no DFT-determined bandgaps have been reported.

In this work, we investigate the viability of α - and ζ - V_2O_5 as secondary battery cathodes in mono- (Li/Na), di- (Mg/Ca), and trivalent (Al) ion applications by use of density functional theory (DFT) [36–42]. To accomplish this, physical structures, bandgaps, density of states (DOS), cell voltages, and diffusion barriers are all calculated with several forms of the exchange-correlation (xc) functional and van der Waals (vdW) correction schemes. Specifically, the local density approximation (LDA) [37], generalized gradient approximation (GGA) by Perdew, Burke, and Ernzerhof (PBE) [43] and the strongly constrained and appropriately normed (SCAN) [44] functionals and the DFT-D3 [45] and rVV10 [46,47] vdW corrections were used. Additionally, the DFT+ U [48,49] method was employed with PBE to accurately account for the on-site Coulomb interactions between the vanadium 3d electrons.

2. Computational details

2.1. Computational methods

DFT calculations were performed using the Vienna *Ab-initio* Simulation Package (VASP) software [50]. VASP utilizes a plane wave basis set in the expansion of the Kohn-Sham single particle wave functions. The ion-electron interactions were approximated using the projector augmented wave (PAW) potentials [51]. Both the kinetic energy cutoff and Γ -centered Monkhorst-Pack [52] k -point grid were tested for convergence against a 1 meV/atom threshold. It was determined that a 700 eV kinetic energy cutoff and a $9 \times 9 \times 3 / 3 \times 15 \times 9$ k -point grid for the α/ζ unit cell was sufficient. Appropriate smearing methods for sampling the Brillouin zone, such as the Methfessel-Paxton [53] smearing method for the intercalated structures, were employed. All structural relaxations and energy calculations were performed using the conjugate-gradient algorithm [54] and the ground state configuration was considered reached upon net forces on atoms falling below 0.01 eV/Å. Each system was modeled and analyzed using the visualization for electronic and structural analysis (VESTA) software [55].

2.2. Relaxations

Full structural relaxations (including lattice vectors) were completed

on both the α and ζ phases of V_2O_5 using the PBE and PBE-D3 xc-vdW approximations. Concentrations of $x = 0.67$, $x = 0.33$, and $x = 0.20$ in $M_xV_2O_5$ ($M = \text{Li, Na, Mg, Ca, Al}$) were used for intercalations of mono-, di-, and trivalent ions, respectively. These concentrations were selected to result in approximately the same number of electrons being transferred between the different valencies of each species.

Intercalation of Na in both V_2O_5 phases was completed through a series of site testing, with a total of 18 sites spanning the unit cell. All combinations of ionic sites for each concentration were tested to determine the most energetically favorable configuration of ionic positions. These favorable positions were used as the basis for testing in the subsequent concentration. Completion of these calculations was done to construct a full voltage profile of Na in both α - and ζ - V_2O_5 .

2.3. Density of states

All xc functionals and vdW corrections, as listed in Sec. 1, were implemented in the DOS calculations on the empty α - and ζ - V_2O_5 structures. However, only PBE and PBE-D3 were utilized on intercalated structures. While it is widely accepted that standard DFT underestimates bandgaps [56] and that a + U correction improves on this estimation [14,15,28,57], others have demonstrated that PBE is of sufficient accuracy in determining the bandgap on such V_2O_5 systems [25,30]. Therefore, the + U correction was not included in our intercalated electronic investigations. Further justification for excluding a + U method is provided in Sec. 2.5. For all DOS calculations, the non-spherical contributions from gradient corrections inside the PAW spheres were included.

2.4. Voltage

The voltage was found for intercalation of a single ion as the difference in total energies of the different constituents as

$$V = - \left(\frac{E_{\text{intercalated}} - E_{\text{empty}} - E_{\text{metal}}}{N} \right), \quad (1)$$

where $E_{\text{intercalated}}$ is the energy of the intercalated cathode system, E_{empty} is the energy of the empty cathode, E_{metal} is the energy per atom of a pure metal anode, and N is the valency of the intercalated cation. It should be noted that these calculations are based solely on the internal energy of each constituent; the Gibbs free energy, which requires the consideration of volume and entropy effects, would contribute less than 0.2 V to the calculated voltage and significantly increase the computational expense of calculations [58–64]. Additionally, surface interactions are not considered in this work; it is assumed that ions can readily intercalate into the bulk cathode.

For calculating voltage from intercalating additional ions, the voltage was calculated as:

$$V = - \left(\frac{E_n - E_m - (n - m)E_{\text{metal}}}{(n - m)N} \right). \quad (2)$$

Here, E_n and E_m are the energies of the cathode material with n and m ions intercalated, respectively, where $n > m$. The product in the denominator equates to the total number of electrons transferred during the intercalation from m to n ions. This process was utilized for those intercalation configurations residing on the convex hull plot (See Figs. S5 & S15).

2.5. Nudged elastic band

Ion migration and diffusion barriers were investigated for both Li and Na using the nudged elastic band (NEB) method [65–72]. Standard NEB was chosen over the climbing-image (CI) method [73] as it has been shown to yield results in agreement with the CI method on similar V_2O_5 systems, while saving on computational costs [14]. The + U correction

method was not employed as it can be difficult to converge in NEB [74] and several studies have concluded that the use of a $+U$ correction shows no conclusive evidence of better predicting cation migration [75–79]. Therefore only PBE-D3 was employed in the NEB calculations, which were completed using the results of the Na site testing mentioned in Sec. 2.2. The use of this method allowed migration of ions along the same path and resulting diffusion barriers to be directly dependent on the migrated ion species. A $4 \times 1 \times 2$ supercell for the α phase and $1 \times 2 \times 2$ supercell for the ζ phase were created in an attempt to balance the accuracy of a larger supercell with the corresponding computational cost. Also, supercells were chosen to allow the NEB pathway to start and end at an equivalent site (i.e., the lowest energy position of a single Na ion). Pathways along the a , b , and c lattice vector directions were individually tested, with only the lowest energy pathway presented here. The other pathways and corresponding data are shown in Sec. S2.4 and S4.4 of the Supplemental Information. A total of 3 images for both V_2O_5 phases were used to sample along the migration path. All NEB calculations were performed using a quasi-Newton algorithm [80] and a $0.05 \text{ eV}/\text{\AA}$ criterion for forces on atoms.

3. Results

α - V_2O_5 Empty Properties

3.1. Physical

All structural relaxations of unintercalated α - V_2O_5 resulted in an orthorhombic unit cell, as shown in Fig. 1a, regardless of method (functional). The resulting lattice parameters are shown in Table 1, along with those from experimental works [23,81–83]. The PBE-D3 xc-vdW combination yields better overall agreement with experimental results, particularly in the direction of vdW interactions (i.e., the b lattice parameter). It should be noted, however, that both PBE and PBE-D3 are in good agreement with experimental works while following the trend of DFT to typically overestimate lattice parameters [15, 26–28]. There are no discernible distortions to the VO_5 polyhedra, matching the structure of pristine α - V_2O_5 . Numerous other computational studies have considered α - V_2O_5 and the results of those using similar methods to those employed here are in agreement with our structure (see Sec. S1.1).

3.2. Electronic

Electronic structure calculations were completed using various functionals as described in Sec. 2.3, with resulting bandgaps shown in Fig. 1b. The range of experimental works [30,84–90] is included as a shaded region in Fig. 1b. All tested functionals, with the exception of PBE+ U ($U=8 \text{ eV}$) and PBE-D3, resulted in a bandgap within this experimental range. However, we show that standard DFT falls within the lower end of this range and therefore does not fully contradict the previously mentioned behavior of underestimating bandgaps. Several other DFT-determined bandgaps have been reported and are listed in Table S2, along with the mentioned experimental values, and resulting DOS of the current work are shown in Fig. S1.

α - V_2O_5 Intercalated Properties

3.3. Physical & electronic

Structural relaxations for ion intercalations, as discussed in Sec. 2.2, maintained the orthorhombic unit cell shown in Fig. 1a. Through all intercalations, lattice parameters a , b , and cell volume are larger than those of empty α - V_2O_5 , while c is smaller. The only exceptions were lattice parameter a for the PBE intercalations of $Li_{0.67}$, $Mg_{0.33}$, and $Al_{0.20}$. A more detailed comparison between intercalated lattice parameters is discussed in Sec. S2.1 and all resulting lattice parameters are shown in

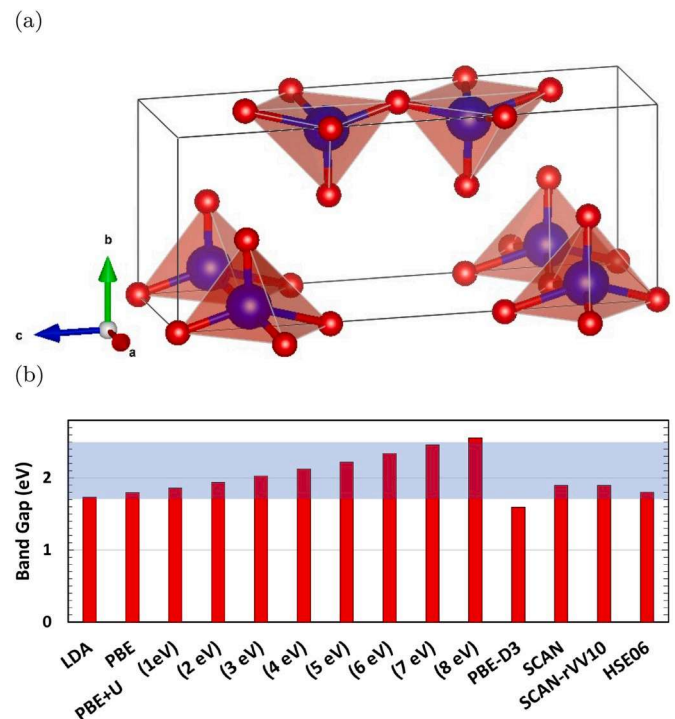


Fig. 1. (a) The unit cell of α - V_2O_5 used in this investigation. Vanadium atoms are shown in blue and oxygen in red. (b) Calculated bandgaps for α - V_2O_5 using various xc-functionals and vdW corrections. Shaded region shows range of experimental values mentioned in Sec. 3.2. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1

Lattice parameters for α - V_2O_5 from this work^a and experimental works. Many DFT studies have also been published and the corresponding data are included in Table S1.

Type of work	Lattice Parameter			Ref.
	$a(\text{\AA})$	$b(\text{\AA})$	$c(\text{\AA})$	
DFT (PBE)	3.567	4.703	11.550	^a
DFT (PBE-D3)	3.553	4.378	11.638	^a
Exp.	3.559	4.367	11.508	[23]
Exp.	3.550	4.360	11.480	[81]
Exp.	3.564	4.373	11.519	[82]
Exp.	3.564	4.368	11.512	[83]

Table S3.

Electronic calculations were completed on the various intercalants at various concentrations (i.e., those shown in Table S3). The intercalation of ions into α - V_2O_5 predominantly results in the filling of vanadium d -states. Increasing the Li concentration fills more of these states, as expected. All systems become metallic upon intercalation. A major shift in the valence and conduction bands occurs upon intercalation, with continual shifting in both with increased Li concentration. All intercalations maintain the split-off conduction band, which is also shown in Refs. [10,30,31,91]. The Li and MV intercalated DOS are shown in Figs. S2 & S3, respectively.

3.4. Voltage

Using Eqns. (1) & (2), voltages were calculated for Li, Na, Mg, Ca and Al in α - V_2O_5 , with results tabulated in Table 2. The concentration of $Al_{0.20}$ is unfavorable (i.e., a negative voltage) and the cause for its exclusion. $Al_{0.10}$, however, is favorable and gave 1.27 V when using PBE and 1.21 V when using PBE-D3. This is of no surprise as others have

reported issues/disadvantages with Al in vanadium oxide secondary batteries [10,14,35,92]. Several experimental [9,19,20,22,24,29,33] and computational [10,13–15,21,26,32] works have reported voltages for the ions investigated here, most of which differ in concentrations of intercalants. This makes direct comparison impractical, however interpolation between these concentrations was completed. When compared to experiment, Li [19,20,22] voltages were underestimated by ~ 0.35 V across the board. Na [19,29] and Mg [24,33] overestimated (~ 0.60 V) with PBE but were at a more reasonable value (within 0.15 V) with PBE-D3. Ca [9] results appear to agree with experiment, but one data point of a different concentration precludes detailed comparison. No experimental values were found for Al intercalation. A comparison to computational data for all ions is examined in Sec. S2.3 and all data referenced here can be found in Table S4.

Additionally, a full voltage profile was calculated for Na intercalation. The final profile is shown in Fig. 2, along with the results from experimental [19,29] and computational [10,13,14,26] investigations. Extrapolating the data from Skaarup et al. [19] (red circles), one can see that DFT noticeably overestimates the voltage for Na in α -V₂O₅. However, close agreement (within 0.20 V) is achieved when compared to the work by Ali et al. [29] (orange circles). Comparison of Fig. 2 with our calculated Li concentrations reveals that Na initially provides a higher voltage (3.39 V versus 3.22 V). This initial Li voltage is represented by a red x on Fig. 7. These two ions provide similar voltages until a concentration of $\sim x = 0.67$, where Li yields 2.86 V and Na falls between 3.28 V ($x = 0.50$) and 2.79 V ($x = 0.75$). The same behavior can be seen when compared to experimental data for Li [19,20,22]. The raw data (Fig. S4) and corresponding convex hull plot (Fig. S5) for our Na voltage profile are discussed in Sec. S2.3.

3.5. Nudged elastic band

NEB calculations were carried out as described in Sec. 2.5 for Li and Na ions. The lowest energy path of ion migration is shown in Fig. 3a and corresponding energy barriers in Fig. 3b. While it is known that DFT has a tendency to overestimate diffusion barriers [93,94], several other computational studies [10,14,25–27,31] investigated migration barriers for these ions as well. Their reported barriers range from 0.12 to 0.35 eV for Li and 0.85 to 1.16 eV for Na, while our barriers are 0.39 eV for Li and 1.27 eV for Na. Although our barriers are higher than others (as expected with previously mentioned behavior of DFT), they are in close agreement with their work. According to the 0.525 eV threshold determined by Rong et al. [35], our determined Li barrier is at a level appropriate for application in a secondary battery. Na, however, is well over this threshold and would hinder its applicability in ambient conditions, despite its high voltage. The cause for this larger barrier may be twofold: the larger ionic radius of Na and the reported slow kinetics of α -V₂O₅ [11,30,95–99]. All barriers discussed above are included in Table S5. As a result of this sluggish behavior, it has been proposed to use a different VO polymorph [12,30,34,100] and introduces the second half of this investigation.

Table 2

Calculated voltage for the various M_x concentrations in α -V₂O₅. NOTE: The Na voltage reported here does not correspond to the voltage profile shown in Fig. 2 due to different simulation cell used.

M_x	Voltage (V)	
	PBE	PBE-D3
Li _{0.17}	3.01	2.98
Li _{0.33}	3.16	3.39
Li _{0.50}	2.97	2.98
Li _{0.67}	2.86	2.89
Na _{0.67}	2.88	2.49
Mg _{0.33}	2.04	2.05
Ca _{0.33}	2.95	2.95

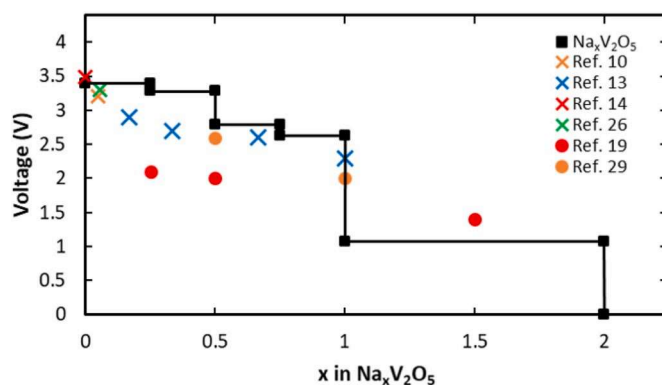


Fig. 2. Voltage profile for Na in α -V₂O₅ when using PBE. Experimental (●) and computational (x) works are also included.

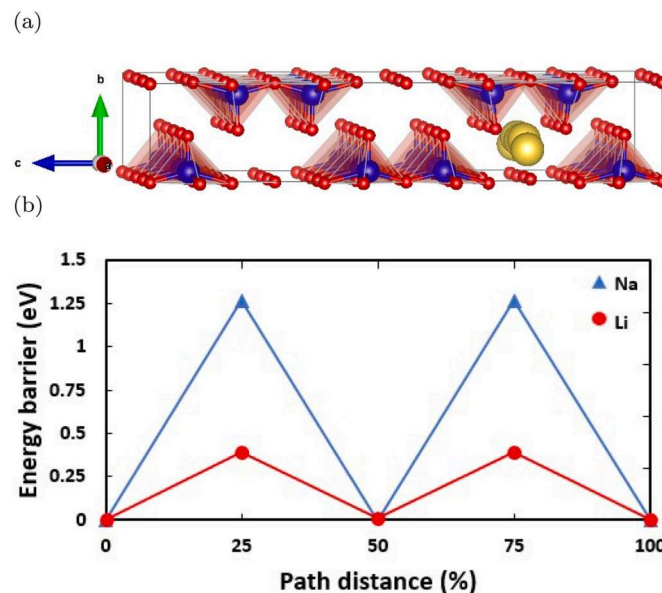


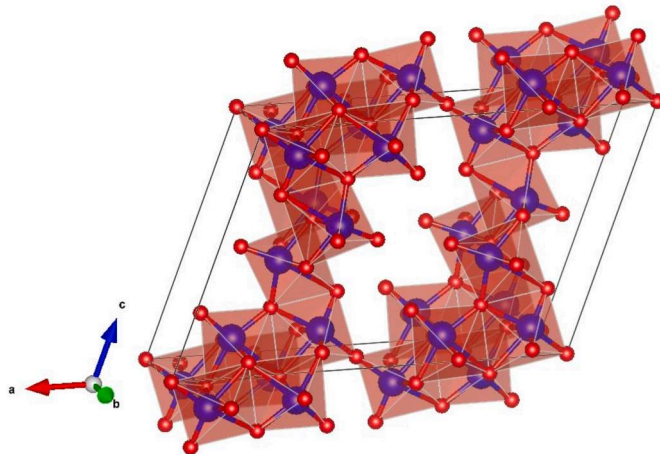
Fig. 3. The (a) lowest energy pathway and (b) corresponding ion migration energy barriers for Li and Na in α -V₂O₅ when using PBE-D3. NOTE: Additional views of the migration pathway can be found in Figs. S7a–c (Sec. S2.4) of the Supplemental Information.

ζ -V₂O₅ Empty Properties

3.6. Physical

The structural relaxations for empty ζ -V₂O₅ resulted in the monoclinic unit cell shown in Fig. 4a for both PBE and PBE-D3. The resulting lattice parameters are shown in Table 3 together with those from experimental works [100–104]. It should be noted that the experimental works cited are not for an empty structure, but instead have traces of intercalants present. These trace intercalants are required to stabilize the ζ -V₂O₅ structure as the empty structure has yet to be successfully synthesized in bulk form [104]. These works had traces of Ag [100,104], Cu [101–103,105], Li [105], or Mg [104] in their structure. The PBE-D3 xc-vdW approximation yields better agreement with experimental results, although PBE is also in good agreement, and all final structures had no observable distortions to the VO₆ octahedra. One other computational work [14] that reported lattice parameters for ζ -V₂O₅ was found in the literature (see Sec. S3.1), but it employed different methods than the ones in the present work.

(a)



(b)

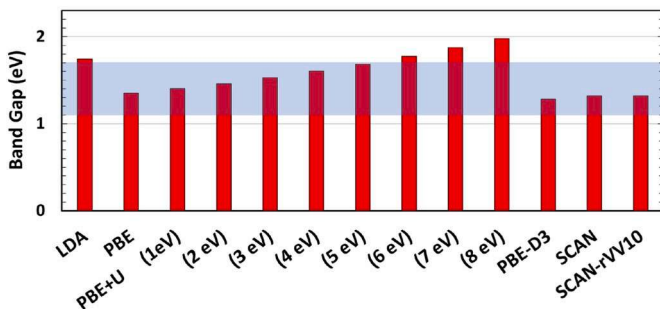


Fig. 4. (a) The unit cell of ζ -V₂O₅ used in this investigation. Vanadium atoms are shown in blue and oxygen in red. (b) Calculated bandgaps for ζ -V₂O₅ using various xc-functionals and vdW corrections. Shaded region shows range of experimental values mentioned in Sec. 3.7. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.7. Electronic

Following the procedure outlined in Sec. 2.3, electronic structure calculations were completed for each intercalation concentration listed in Sec. 2.2 and the resulting bandgaps are shown in Fig. 4b. A shaded region is included to denote the range of reported bandgaps from experimental works [30,100]. These works found a bandgap of 1.50 ± 0.20 eV and 1.1 eV, respectively. Calculations with the functionals of PBE, PBE+ U ($U = 1$ –5 eV), PBE-D3, SCAN and SCAN-rVV10 resulted in bandgaps within this experimental range. To the best of the authors' knowledge, no DFT-determined bandgaps have been reported for ζ -V₂O₅ for further comparison. The DOS plots for this investigation are shown in Fig. S10 and all mentioned bandgaps are included in Table S8.

ζ -V₂O₅ Intercalated Properties

3.8. Physical & electronic

All structural relaxations for intercalated ζ -V₂O₅ maintained the monoclinic unit cell shown in Fig. 4a with no obvious trend in lattice parameters detected. However, a detailed comparison between all lattice parameters is done in Sec. S4.1, with lattice parameters shown in Table S9.

Electronic calculations for the various intercalant concentrations shown in Table S9 were completed. Ion intercalation into ζ -V₂O₅ results in predominantly filling of vanadium d -states. Increasing the Li concentration fills more of these states, as expected, and all systems become

Table 3

Lattice parameters for ζ -V₂O₅ from this work^a and experimental works. One DFT study has been published and the corresponding data are included in Table S7.

Type of work	Lattice Parameter				Ref.
	a(Å)	b(Å)	c(Å)	$\beta(^{\circ})$	
DFT (PBE)	15.760	3.585	10.305	110.400	^a
DFT (PBE-D3)	15.406	3.586	10.192	110.469	^a
Exp.	15.275	3.604	10.098	110.022	[100]
Exp.	15.250	3.620	10.100	106.900	[101]
Exp.	15.233	3.613	10.093	107.300	[102]
Exp.	15.201	3.673	10.093	106.310	[103]
Exp.	15.336	3.612	10.083	109.762	[104]
Exp.	15.375	3.600	10.095	110.300	[105]
Exp.	15.250	3.624	10.205	106.900	[105]

metallic upon ion intercalation. A major shift in the valence and conduction band occurs upon intercalation, with continual shifting in both with increased Li concentration. All Li and MV intercalated DOS are shown in Figs. S11 & S12, respectively.

3.9. Voltage

Voltages for Li, Na, Mg, Ca and Al intercalations in ζ -V₂O₅ were calculated using Eqns. (1) and (2) and are recorded in Table 4. All concentrations were of favorable results (i.e., positive voltages) for PBE and PBE-D3. Few other works have considered ζ -V₂O₅ in the context of this investigation. For example, the work by Liang et al. [106] studied the ζ -phase for lithium ion batteries. However, the structure of their cathode was not pure ζ -V₂O₅ but the sodiated version (β -Na_{0.33}V₂O₅), as was also the case for Kim et al. [107]. Nevertheless, comparison is made by interpolation between data points as no direct match exists between concentrations used. PBE resulted in voltages inline with those reported by West et al. [20] in their experimental work, while PBE-D3 gave noticeably higher voltages. The Li_{0.67} concentration was anomalous for both functionals as it increased from preceeding concentrations with PBE and experienced a considerable drop with PBE-D3. The data referenced here are included in Table S10.

Beyond the intercalations presented in Table 4, a full voltage profile (Fig. 5) was completed for Na in ζ -V₂O₅. An experimental [8] and computational [14] work also reported voltage values for Na, which are included in Fig. 5. It should be noted that intercalation was anomalous for both functionals as it increased from preceeding concentrations with PBE and experienced a considerable drop with PBE-D3. The data referenced here are included in Table S10.

Table 4

Calculated voltage for the various M_x concentrations in ζ -V₂O₅. NOTE: The Na voltage reported here does not correspond to the voltage profile shown in Fig. 5 due to different simulation cell used.

M _x	Voltage (V)	
	PBE	PBE-D3
Li _{0.17}	3.61	3.99
Li _{0.33}	3.55	3.84
Li _{0.50}	3.37	3.59
Li _{0.67}	3.50	2.84
Na _{0.67}	3.26	3.69
Mg _{0.33}	2.11	2.18
Ca _{0.33}	2.82	2.91
Al _{0.20}	1.48	1.54

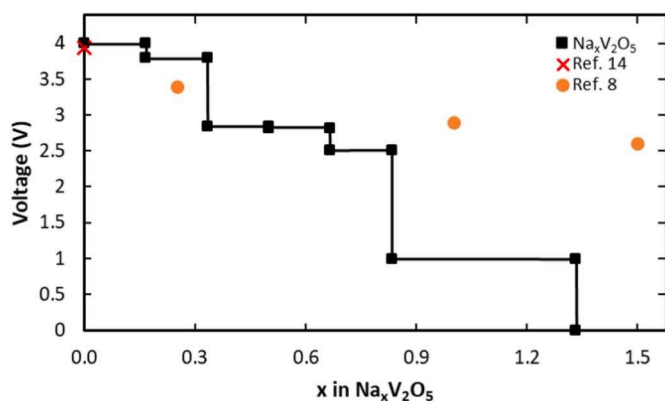


Fig. 5. Voltage profile for Na in ζ -V₂O₅ when using PBE. Experimental (●) and computational (x) works are also included.

for our Na profile are discussed in Sec. S4.3.

3.10. Nudged elastic band

NEB calculations were carried out as described in Sec. 2.5. The lowest energy path of Li and Na ion migration is shown in Fig. 6a and corresponding energy barriers in Fig. 6b. One other computational study investigated migration barriers for Li and Na in ζ -V₂O₅, to the best of the authors' knowledge. This work by Parija et al. [14] found a barrier of ~ 0.11 eV for Li and ~ 0.38 eV for Na. The present work, however, found

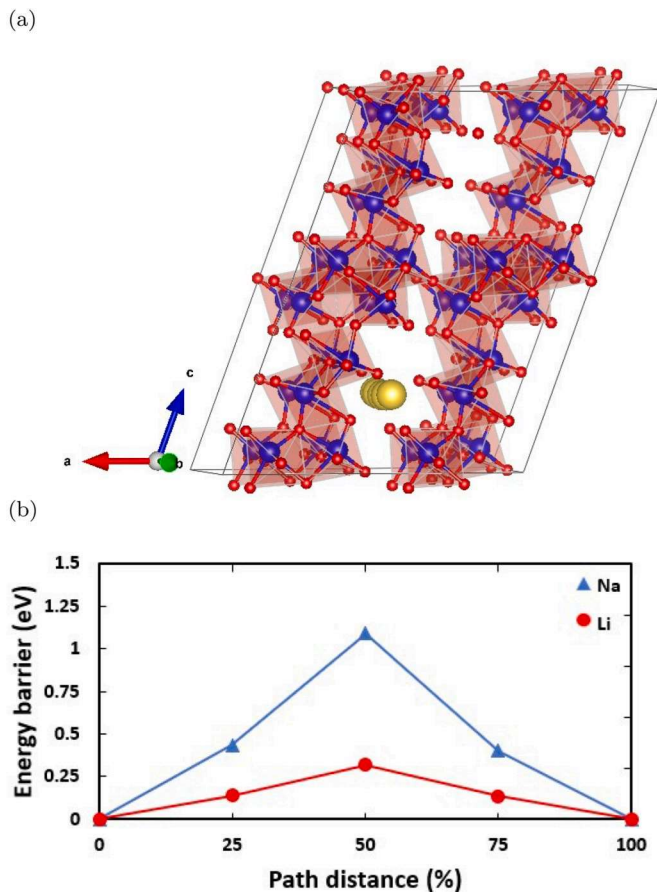


Fig. 6. The (a) lowest energy pathway and (b) corresponding ion migration energy barriers for Li and Na in ζ -V₂O₅ when using PBE-D3. NOTE: Additional views of the migration pathway can be found in Figs. S16a–c (Sec. S4.4) of the Supplemental Information.

barriers of 0.32 eV for Li and 1.09 eV for Na. Explicit comparison of barriers between the present work and that of Parija et al. would be imprudent due to different methods employed to calculate each barrier. Comparing our barriers to the 0.525 eV threshold determined by Rong et al. [35], ζ -V₂O₅ is capable of LIB application based on calculated Li ion diffusion kinetics. However, the high Na ion diffusion barrier would disqualify ζ -V₂O₅ from a Na ion battery (NIB) application. All barriers mentioned above can be found in Table S11.

4. Discussion

The majority of the ion intercalations discussed in Sec. 2.2 result in higher voltages in ζ -V₂O₅ over α -V₂O₅. Li_{0.67} is 0.05 V lower in ζ when using the PBE-D3 functional and Ca_{0.33} is 0.04 V lower. For PBE, however, the voltage for Ca_{0.33} is 0.13 V lower in ζ . It is suspected that due to the structural changes ζ -V₂O₅ underwent at this concentration (and functional), the structure became more unfavorable and explains the loss in voltage. For Ca_{0.33} in ζ , it deviates the most from the empty structure in lattice parameter *a* and *b*. It also has the second largest change in *c* and volume. Another possibility may be due to unfavorable ionic positions as site testing was not completed for intercalations reported in Tables 2 & 4.

For similar charge intercalation (i.e., Li_{0.67}, Na_{0.67}, Mg_{0.33}, Ca_{0.33}, and Al_{0.20}), voltages can vary widely. For PBE in α , the ions rank Ca > Na > Li > Mg > Al. This trend follows largest to smallest ion radius. PBE-D3 ranks the same with Li and Na switched. In either case, Li, Na, and Ca provide quite similar voltages. However, both Li_{0.67} and Na_{0.67} result in over a 10% change in volume with both functionals. In ζ -V₂O₅, the ranking for similar charge intercalation changes dramatically whether PBE or PBE-D3 is used. A constant between them is that Mg and Al rank as the lowest two in both polymorphs of V₂O₅ and all intercalations result in a percent change in volume less than 4.2%, with some resulting in a contraction. Such changes in volume are desirable as they suggest a low chance of cathode cracking compared to systems with higher volume changes.

A complete voltage profile was obtained for the α - and ζ -V₂O₅ phases, which are combined in Fig. 7. The ζ phase starts with a higher voltage than α (4.0 V versus 3.4 V) and the profiles are quite similar in behavior up to the concentration of $\sim x = 1.0$, where both experience a large drop in voltage (~ 1.5 V). Beyond this concentration, ζ -V₂O₅ intercalates Na up to $x = 1.33$ in Na_xV₂O₅, while α continues to $x = 2.0$. Regarding utility as a NIB, ζ -V₂O₅ is on par with α -V₂O₅ in voltage but lacks the intercalation capacity. At maximum intercalation, α -V₂O₅ yields a specific capacity of 235 mAhg⁻¹ and ζ -V₂O₅ yields 168 mAhg⁻¹.

Diffusion barriers for Li and Na were calculated for both phases of V₂O₅. The α phase resulted in a barrier of 0.39 eV for Li and 1.27 eV for Na. However, the ζ phase resulted in barriers of 0.32 eV and 1.09 eV,

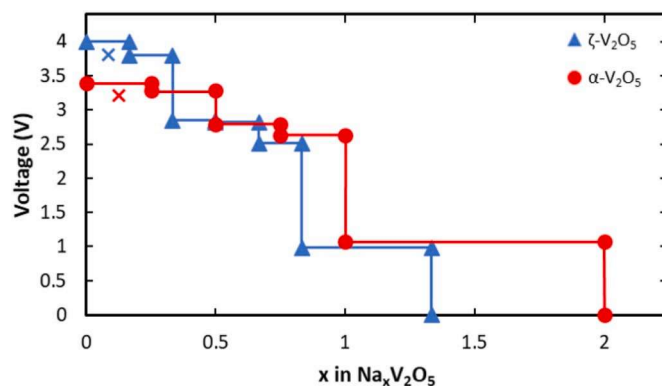


Fig. 7. Voltage profile for Na in α - and ζ -V₂O₅. The x's correspond to replacing the first intercalated Na with Li to compare initial voltages between the two ions.

respectively. The lower barriers of ζ -V₂O₅ are surprising due to the rigid tunneled structure of ζ -V₂O₅. One would expect the larger intercalation channels and more easily manipulated layered structure (e.g., layer separation) of α -V₂O₅ would lead to lower migration barriers. As a result of these barriers, α - and ζ -V₂O₅ show no feasible operation in NIBs, though a LIB functionality is feasible.

5. Conclusions

The energetic and structural properties of mono- (Li/Na), di- (Mg/Ca), and trivalent (Al) intercalations in α - and ζ -V₂O₅ systems were studied using various combinations of exchange-correlation functionals and van der Waals corrections. When comparing similar charge intercalation (i.e., Li_{0.67}, Na_{0.67}, Mg_{0.33}, Ca_{0.33}, and Al_{0.20}) in each phase, Ca provided the highest voltage (2.95 V) in α . Li provided the highest in ζ when using PBE (3.50 V) and Na was the highest with PBE-D3 (3.69 V). However, Li, Na, and Ca all provide similar voltages in each phase for a given xc functional. Other than Al_{0.20} in α -V₂O₅, all intercalation concentrations investigated resulted in positive voltage values. Full voltage profiles were completed for Na intercalation in each phase using PBE-D3. ζ -V₂O₅ had a higher initial voltage than that of α (4.00 V vs. 3.4 V), but lacks the overall specific capacity (168 mAhg⁻¹ vs. 235 mAhg⁻¹). Diffusion barriers for Li and Na ion migrations were also investigated. Each ion had a barrier of 0.39/0.32 eV and 1.27/1.09 eV in α / ζ -V₂O₅, respectively. With the higher diffusion barrier and lower total specific capacity, ζ -V₂O₅ is not an ideal sodium ion battery cathode material. However, both α - and ζ -V₂O₅ possess voltages and diffusion barriers conducive to Li ion battery applications. While α - and ζ -V₂O₅ are not ideal candidates for Na ion batteries, several other vanadium oxide polymorphs exist and can be studied for such applications.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2020.228096>.

References

- M.S. Whittingham, Electrical energy storage and intercalation chemistry, *Science* 192 (4244) (1976) 1126–1127, <https://doi.org/10.1126/science.192.4244.1126>.
- H. Ikeda, T. Saito, H. Tamaru, Lithium-Manganese dioxide cell (1), *Denki Kagaku* 45 (5) (1977) 314–318.
- H. Ikeda, S. Uena, T. Saito, S. Nakaido, H. Tamaru, Lithium-Manganese dioxide cell (2), *Denki Kagaku* 45 (6) (1977) 391–395.
- K. Mizushima, P. Jones, P. Wiseman, J. Goodenough, Li_xCoO₂ (0 < x < 1): a new cathode material for batteries of high energy density, *Mater. Res. Bull.* 15 (6) (1980) 783–789, [https://doi.org/10.1016/0025-5408\(80\)90012-4](https://doi.org/10.1016/0025-5408(80)90012-4).
- J. Speirs, M. Contestabile, The future of lithium availability for electric vehicle batteries, *Renew. Sustain. Energy Rev.* 35 (2014) 183–193, <https://doi.org/10.1016/j.rser.2014.04.018>.
- P.W. Gruber, P.A. Medina, G.A. Keoleian, S.E. Kesler, M.P. Everson, T. J. Wallington, Global lithium availability: a constraint for electric vehicles? *J. Ind. Ecol.* 15 (5) (2011) 760–775, <https://doi.org/10.1111/j.1530-9290.2011.00359.x>.
- M.S. Whittingham, Lithium batteries and cathode materials, *Chem. Rev.* 104 (10) (2004) 4271–4301, <https://doi.org/10.1021/cr020731c>.
- R.D. Apostolova, E.M. Shembel, V.M. Nagirnyi, D. Aurbach, B. Markovsky, Y. Langzam, Synthesis and examination of electrolytic sodium-vanadium oxide compounds intended for cathodes of lithium batteries: the mechanism of formation of electrolytic bronze β -Na_xV₂O₅, *Russ. J. Electrochem.* 37 (10) (2001) 1041–1049, <https://doi.org/10.1023/A:1012314912411>.
- G.G. Amatucci, F. Badway, A. Singhal, B. Beaudoin, G. Skandan, T. Bowmer, I. Plitz, N. Pereira, T. Chapman, R. Jaworski, Investigation of yttrium and polyvalent ion intercalation into nanocrystalline vanadium oxide, *J. Electrochem. Soc.* 148 (8) (2002) A940–A950, <https://doi.org/10.1149/1.1383777>.
- V.V. Kulish, S. Manzhos, Comparison of Li, Na, Mg and Al-ion insertion in vanadium pentoxides and vanadium dioxides, *RSC Adv.* 7 (2017) 18643–18649, <https://doi.org/10.1039/c7ra02474f>.
- P.M. Marley, S. Banerjee, Reversible interconversion of a divalent vanadium bronze between δ and β quasi-1D structures, *Inorg. Chem.* 51 (9) (2012) 5264–5269, <https://doi.org/10.1021/ic300191y>.
- P.M. Marley, G.A. Horrocks, K.E. Pelcher, S. Banerjee, Transformers: the changing phases of low-dimensional vanadium oxide bronzes, *Chem. Commun.* 51 (25) (2015) 5181–5198, <https://doi.org/10.1039/c4cc08673b>.
- D. Wang, H. Liu, J.D. Elliott, L.M. Liu, W.M. Lau, Robust vanadium pentoxide electrodes for sodium and calcium ion batteries: thermodynamic and diffusion mechanical insights, *J. Mater. Chem. A* 4 (32) (2016) 12516–12525, <https://doi.org/10.1039/c6ta03595g>.
- A. Parija, Y. Liang, J.L. Andrews, L.R. De Jesus, D. Prendergast, S. Banerjee, Sonochemically de-intercalated phases of V₂O₅ as cathode materials for multivalent intercalation batteries: a first-principles evaluation, *Chem. Mater.* 28 (16) (2016) 5611–5620, <https://doi.org/10.1021/acs.chemmater.6b01006>.
- A. Parija, D. Prendergast, S. Banerjee, Evaluation of multivalent cation insertion in single- and double-layered polymorphs of V₂O₅, *ACS Appl. Mater. Interfaces* 9 (28) (2017) 23756–23765, <https://doi.org/10.1021/acsami.7b05556>.
- R. Verrelli, A.P. Black, C. Pattanathummasid, D.S. Tchitchekova, A. Ponrouch, J. Oró-Solé, C. Frontera, F. Bardé, P. Rozier, M.R. Palacín, On the strange case of divalent ions intercalation in V₂O₅, *J. Power Sources* 407 (2018) 162–172, <https://doi.org/10.1016/j.jpowsour.2018.08.024>.
- V. Eyert, K.-H. Höck, Electronic structure of V₂O₅: role of octahedral deformations, *Phys. Rev. B* 57 (20) (1998) 12727–12737, <https://doi.org/10.1103/PhysRevB.57.12727>.
- N.A. Chernova, M. Roppolo, A.C. Dillon, M.S. Whittingham, Layered vanadium and molybdenum oxides: batteries and electrochromics, *J. Mater. Chem.* 19 (17) (2009) 2526–2552, <https://doi.org/10.1039/b819629j>.
- S. Skaarup, K. West, B. Zachau-Christiansen, T. Jacobsen, Solid state sodium batteries, in: T. Takahashi (Ed.), *High Conductivity Solid Ionic Compounds: Recent Trends and Applications*, World Scientific, Singapore, 1988, pp. 47–58.
- K. West, B. Zachau-Christiansen, T. Jacobsen, S. Skaarup, Lithium insertion into vanadium pentoxide bronzes, *Solid State Ionics* 76 (1–2) (1995) 15–21, [https://doi.org/10.1016/0167-2738\(95\)90437-M](https://doi.org/10.1016/0167-2738(95)90437-M).
- J.S. Braithwaite, C.R.A. Catlow, J.D. Gale, J.H. Harding, Lithium intercalation into vanadium pentoxide: a theoretical study, *Chem. Mater.* 11 (8) (1999) 1990–1998, <https://doi.org/10.1021/cm980735r>.
- C. Leger, S. Bach, P. Soudan, J.-P. Pereira-Ramos, Structural and electrochemical properties of ω -Li_xV₂O₅ (0.4 ≤ x ≤ 3) as rechargeable cathodic material for lithium batteries, *J. Electrochem. Soc.* 152 (1) (2005) A236, <https://doi.org/10.1149/1.1836155>.
- S.L. Chou, J.Z. Wang, J.Z. Sun, D. Wexler, M. Forsyth, H.K. Liu, D.R. MacFarlane, S.X. Dou, High capacity, safety, and enhanced cyclability of lithium metal battery using a V₂O₅ nanomaterial cathode and room temperature ionic liquid electrolyte, *Chem. Mater.* 20 (22) (2008) 7044–7051, <https://doi.org/10.1021/cm801468q>.
- G. Gershinsky, H.D. Yoo, Y. Gofer, D. Aurbach, Electrochemical and spectroscopic analysis of Mg²⁺ intercalation into thin film electrodes of layered oxides: V₂O₅ and MoO₃, *Langmuir* 29 (34) (2013) 10964–10972, <https://doi.org/10.1021/la402391f>.
- B. Zhou, H. Shi, R. Cao, X. Zhang, Z. Jiang, Theoretical study on the initial stage of a magnesium battery based on a V₂O₅ cathode, *Phys. Chem. Chem. Phys.* 16 (34) (2014) 18578–18585, <https://doi.org/10.1039/c4cp02230k>.
- J. Carrasco, Role of van der Waals forces in thermodynamics and kinetics of layered transition metal oxide electrodes: alkali and alkaline-earth ion insertion into V₂O₅, *J. Phys. Chem. C* 118 (34) (2014) 19599–19607, <https://doi.org/10.1021/jp505821w>.
- G.S. Gautam, P. Canepa, R. Malik, M. Liu, K. Persson, G. Ceder, First-principles evaluation of multi-valent cation insertion into orthorhombic V₂O₅, *Chem. Commun.* 51 (2015) 13619–13622, <https://doi.org/10.1039/c5cc04947d>.
- D.O. Scanlon, A. Walsh, B.J. Morgan, G.W. Watson, An ab initio study of reduction of V₂O₅ through the formation of oxygen vacancies and Li intercalation, *J. Phys. Chem. C* 112 (26) (2008) 9903–9911, <https://doi.org/10.1021/jp711334f>.
- G. Ali, J.H. Lee, S.H. Oh, B.W. Cho, K.W. Nam, K.Y. Chung, Investigation of the Na intercalation mechanism into nanosized V₂O₅/C composite cathode material for Na-ion batteries, *ACS Appl. Mater. Interfaces* 8 (9) (2016) 6032–6039, <https://doi.org/10.1021/acsami.5b11954>.
- T.M. Tolhurst, L. Leedahl, J.L. Andrews, P.M. Marley, S. Banerjee, A. Moewes, Contrasting 1D tunnel-structured and 2D layered polymorphs of V₂O₅: relating crystal structure and bonding to band gaps and electronic structure, *Phys. Chem. Chem. Phys.* 18 (23) (2016) 15798–15806, <https://doi.org/10.1039/c6cp02096h>.
- L.R. De Jesus, G.A. Horrocks, Y. Liang, A. Parija, C. Jaye, L. Wangoh, J. Wang, D. A. Fischer, L.F. Piper, D. Prendergast, S. Banerjee, Mapping polaronic states and lithiation gradients in individual V₂O₅ nanowires, *Nat. Commun.* 7 (2016), 12022, <https://doi.org/10.1038/ncomms12022>.
- A. Parija, G.R. Waetzig, J.L. Andrews, S. Banerjee, Traversing energy landscapes away from equilibrium: strategies for accessing and utilizing metastable phase space, *J. Phys. Chem. C* 122 (45) (2018) 25709–25728, <https://doi.org/10.1021/acs.jpcc.8b04622>.
- H.D. Yoo, J.R. Jokisaari, Y.-S. Yu, B.J. Kwon, L. Hu, S. Kim, S.-D. Han, M. Lopez, S.H. Lapidus, G.M. Nolis, B.J. Ingram, I.L. Bolotin, S. Ahmed, R.F. Klie, J. T. Vaughney, T.T. Fister, J. Cabana, Intercalation of magnesium into a layered

- vanadium oxide with high capacity, *ACS Energy Lett.* 4 (7) (2019) 1528–1534, <https://doi.org/10.1021/acseenergylett.9b00788>.
- [34] V.P. Filonenko, M. Sundberg, P.E. Werner, I.P. Zibrov, Structure of a high-pressure phase of vanadium pentoxide, β -V₂O₅, *Acta Crystallogr. B* 60 (4) (2004) 375–381, <https://doi.org/10.1107/S0108768104012881>.
- [35] Z. Rong, R. Malik, P. Canepa, G. Sai Gautam, M. Liu, A. Jain, K. Persson, G. Ceder, Materials design rules for multivalent ion mobility in intercalation structures, *Chem. Mater.* 27 (17) (2015) 6016–6021, <https://doi.org/10.1021/acs.chemmater.5b02342>.
- [36] P. Hohenberg, W. Kohn, Inhomogeneous electron gas, *Phys. Rev.* 136 (3B) (1964) B864–B871, <https://doi.org/10.1103/PhysRev.136.B864>.
- [37] W. Kohn, L.J. Sham, Self-consistent equations including exchange and correlation effects, *Phys. Rev.* 140 (4A) (1965) A1133–A1138, <https://doi.org/10.1103/PhysRev.140.A1133>.
- [38] J.P. Perdew, A. Zunger, Self-interaction correction to density-functional approximations for many-electron systems, *Phys. Rev. B* 23 (10) (1981) 5048–5079, <https://doi.org/10.1103/PhysRevB.23.5048>.
- [39] S. Lundqvist, N.H. March, *Theory of the Inhomogeneous Electron Gas*, first ed., Plenum, New York, 1983 <https://doi.org/10.1007/978-1-4899-0415-7>.
- [40] W. Kohn, A.D. Becke, R.G. Parr, Density functional theory of electronic structure, *J. Phys. Chem.* 100 (31) (1996) 12974–12980, <https://doi.org/10.1021/jp960669l>.
- [41] W. Koch, M.C. Holthausen, *A Chemist's Guide to Density Functional Theory*, second ed., Wiley-VCH, Weinheim, 2001 <https://doi.org/10.1002/3527600043>.
- [42] K. Capelle, A bird's-eye view of density-functional theory, *Braz. J. Phys.* 36 (4A) (2006) 1318–1348, <https://doi.org/10.1590/S0103-97332006000700035>.
- [43] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* 77 (18) (1996) 3865–3868, <https://doi.org/10.1103/PhysRevLett.77.3865>.
- [44] J. Sun, A. Ruzsinszky, J.P. Perdew, Strongly constrained and appropriately normed semilocal density functional, *Phys. Rev. Lett.* 115 (3) (2015), 036402, <https://doi.org/10.1103/PhysRevLett.115.036402>.
- [45] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, A consistent and accurate *ab initio* parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu, *J. Chem. Phys.* 132 (15) (2010) 154104, <https://doi.org/10.1063/1.3382344>.
- [46] R. Sabatini, T. Gorni, S. De Gironcoli, Nonlocal van der Waals density functional made simple and efficient, *Phys. Rev. B* 87 (4) (2013), 041108, <https://doi.org/10.1103/PhysRevB.87.041108>.
- [47] H. Peng, Z.-H. Yang, J. Sun, J.P. Perdew, Versatile van der Waals density functional based on a meta-generalized gradient approximation, *Phys. Rev. X* 6 (4) (2016), 041005, <https://doi.org/10.1103/PhysRevX.6.041005>.
- [48] V.I. Anisimov, J. Zaanen, O.K. Andersen, Band theory and Mott insulators: Hubbard *U* instead of Stoner *I*, *Phys. Rev. B* 44 (3) (1991) 943–954, <https://doi.org/10.1103/PhysRevB.44.943>.
- [49] S. Dudarev, G. Botton, Electron-energy-loss spectra and the structural stability of nickel oxide: an LSDA+*U* study, *Phys. Rev. B* 57 (3) (1998) 1505–1509, <https://doi.org/10.1103/PhysRevB.57.1505>.
- [50] G. Kresse, J. Furthmüller, Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set, *Phys. Rev. B* 54 (16) (1996) 11169–11186, <https://doi.org/10.1103/PhysRevB.54.11169>.
- [51] P.E. Blöchl, Projector augmented-wave method, *Phys. Rev. B* 50 (24) (1994) 17953–17979, <https://doi.org/10.1103/PhysRevB.50.17953>.
- [52] H.J. Monkhorst, J.D. Pack, Special points for Brillouin-zone integrations, *Phys. Rev. B* 13 (12) (1976) 5188–5192, <https://doi.org/10.1103/PhysRevB.13.5188>.
- [53] M. Methfessel, A.T. Paxton, High-precision sampling for Brillouin-zone integration in metals, *Phys. Rev. B* 40 (6) (1989) 3616–3621, <https://doi.org/10.1103/PhysRevB.40.3616>.
- [54] W.H. Press, S.A. Teukolsky, W.T. Vetterling, B.P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, third ed., Cambridge University Press, New York, 2007 <https://doi.org/10.1145/1874391.187410>.
- [55] K. Momma, F. Izumi, VESTA 3 for three-dimensional visualization of crystal, volumetric and morphology data, *J. Appl. Crystallogr.* 44 (6) (2011) 1272–1276, <https://doi.org/10.1107/S0021889811038970>.
- [56] Y. Zhang, N. Holzwarth, R. Williams, Electronic band structures of the scheelite materials CaMoO₄, CaWO₄, PbMoO₄, and PbWO₄, *Phys. Rev. B* 57 (20) (1998) 12738–12750, <https://doi.org/10.1103/PhysRevB.57.12738>.
- [57] L. Wang, T. Maxisch, G. Ceder, Ceder, Oxidation energies of transition metal oxides within the GGA+*U* framework, *Phys. Rev. B* 73 (19) (2006), 195107, <https://doi.org/10.1103/PhysRevB.73.195107>.
- [58] M.K. Aydinol, G. Ceder, First principles prediction of insertion potentials in LiMn oxides for secondary Li batteries, *J. Electrochem. Soc.* 144 (11) (1997) 3832–3835, <https://doi.org/10.1149/1.1838099>.
- [59] M.K. Aydinol, A.F. Kohan, G. Ceder, Ab initio calculation of the intercalation voltage of lithium-transition-metal oxide electrodes for rechargeable batteries, *J. Power Sources* 68 (2) (1997) 664–668, [https://doi.org/10.1016/S0378-7753\(96\)02638-9](https://doi.org/10.1016/S0378-7753(96)02638-9).
- [60] M. Aydinol, A. Kohan, G. Ceder, K. Cho, J. Joannopoulos, Ab initio study of lithium intercalation in metal oxides and metal dichalcogenides, *Phys. Rev. B* 56 (3) (1997) 1354–1365, <https://doi.org/10.1103/PhysRevB.56.1354>.
- [61] E. Deiss, A. Wokaun, J.-L. Barras, C. Daul, P. Dufek, Average voltage, energy density, and specific energy of lithium-ion batteries, *J. Electrochem. Soc.* 144 (11) (1997) 3877–3881, <https://doi.org/10.1149/1.1838105>.
- [62] S.K. Mishra, G. Ceder, Structural stability of lithium manganese oxides, *Phys. Rev. B* 59 (9) (1999) 6120–6130, <https://doi.org/10.1103/PhysRevB.59.6120>.
- [63] G. Ceder, A. Van Der Ven, Phase diagrams of lithium transition metal oxides: investigations from first principles, *Electrochim. Acta* 45 (1–2) (1999) 131–150, [https://doi.org/10.1016/S0013-4686\(99\)00199-1](https://doi.org/10.1016/S0013-4686(99)00199-1).
- [64] Y.S. Meng, M.E. Arroyo-De Dompablo, First principles computational materials design for energy storage materials in lithium ion batteries, *Energy Environ. Sci.* 2 (2009) 589–609, <https://doi.org/10.1039/b901825e>.
- [65] G. Mills, H. Jónsson, Quantum and thermal effects in H₂ dissociative adsorption: evaluation of free energy barriers in multidimensional quantum systems, *Phys. Rev. Lett.* 72 (7) (1994) 1124–1127, <https://doi.org/10.1103/PhysRevLett.72.1124>.
- [66] G. Mills, H. Jónsson, G.K. Schenter, Reversible work transition state theory: application to dissociative adsorption of hydrogen, *Surf. Sci.* 324 (2–3) (1995) 305–337, [https://doi.org/10.1016/0039-6028\(94\)00731-4](https://doi.org/10.1016/0039-6028(94)00731-4).
- [67] G. Henkelman, H. Jónsson, Improved tangent estimate in the nudged elastic band method for finding minimum energy paths and saddle points, *J. Chem. Phys.* 113 (22) (2000) 9978–9985, <https://doi.org/10.1063/1.1323224>.
- [68] G. Henkelman, H. Jónsson, Theoretical calculations of dissociative adsorption of CH₄ on an Ir(111) surface, *Phys. Rev. Lett.* 86 (4) (2001) 664–667, <https://doi.org/10.1103/PhysRevLett.86.664>.
- [69] G. Henkelman, G. Jóhannesson, H. Jónsson, Methods for finding saddle points and minimum energy paths, in: S.D. Schwartz (Ed.), *Theoretical Methods in Condensed Phase Chemistry*. Progress in Theoretical Chemistry and Physics, vol. 5, Springer, Dordrecht, 2002, pp. 269–302, https://doi.org/10.1007/0-306-46949-9_10.
- [70] D. Sheppard, R. Terrell, G. Henkelman, Optimization methods for finding minimum energy paths, *J. Chem. Phys.* 128 (13) (2008) 134106, <https://doi.org/10.1063/1.2841941>.
- [71] D. Sheppard, G. Henkelman, Paths to which the nudged elastic band converges, *J. Comput. Chem.* 32 (8) (2011) 1769–1771, <https://doi.org/10.1002/jcc.21748>.
- [72] D. Sheppard, P. Xiao, W. Chemelewski, D.D. Johnson, G. Henkelman, A generalized solid-state nudged elastic band method, *J. Chem. Phys.* 136 (7) (2012), 074103, <https://doi.org/10.1063/1.3684549>.
- [73] G. Henkelman, B.P. Uberuaga, H. Jónsson, A climbing image nudged elastic band method for finding saddle points and minimum energy paths, *J. Chem. Phys.* 113 (22) (2000) 9901–9904, <https://doi.org/10.1063/1.1329672>.
- [74] M. Liu, Z. Rong, R. Malik, P. Canepa, A. Jain, G. Ceder, K.A. Persson, Spinel compounds as multivalent battery cathodes: a systematic evaluation based on *ab initio* calculations, *Energy Environ. Sci.* 8 (2015) 964–974, <https://doi.org/10.1039/c4ee03389b>.
- [75] D. Morgan, A. Van der Ven, G. Ceder, Li conductivity in Li_xMPO₄ (M = Mn, Fe, Co, Ni) olivine materials, *Electrochem. Solid State Lett.* 7 (2) (2004) A30–A32, <https://doi.org/10.1149/1.1633511>.
- [76] B. Xu, S. Meng, Factors affecting Li mobility in spinel LiMn₂O₄-A first-principles study by GGA and GGA+*U* methods, *J. Power Sources* 195 (15) (2010) 4971–4976, <https://doi.org/10.1016/j.jpowsour.2010.02.060>.
- [77] S.P. Ong, V.L. Chevrier, G. Hautier, A. Jain, C. Moore, S. Kim, X. Ma, G. Ceder, Voltage, stability and diffusion barrier differences between sodium-ion and lithium-ion intercalation materials, *Energy Environ. Sci.* 4 (9) (2011) 3680–3688, <https://doi.org/10.1039/c1ee01782a>.
- [78] G.K.P. Dathar, D. Sheppard, K.J. Stevenson, G. Henkelman, Calculations of Li-ion diffusion in olivine phosphates, *Chem. Mater.* 23 (17) (2011) 4032–4037, <https://doi.org/10.1021/cm201604g>.
- [79] H. Lin, Y. Wen, C. Zhang, L. Zhang, Y. Huang, B. Shan, R. Chen, A GGA+*U* study of lithium diffusion in vanadium doped LiFePO₄, *Solid State Commun.* 152 (12) (2012) 999–1003, <https://doi.org/10.1016/j.ssc.2012.03.027>.
- [80] P. Pulay, Convergence acceleration of iterative sequences. The case of SCF iteration, *Chem. Phys. Lett.* 73 (2) (1980) 393–398, [https://doi.org/10.1016/0009-2614\(80\)80396-4](https://doi.org/10.1016/0009-2614(80)80396-4).
- [81] Von J.A.A. Ketelaar, Die Kristallstruktur des Vanadinpentoxids, *Z. Kristallogr.* 95 (1–6) (1936) 9–27, <https://doi.org/10.1524/zkri.1936.95.1.9>.
- [82] A. Byström, K.-A. Wilhelm, O. Brotzen, Vanadium pentoxide - a compound with five-coordinated vanadium atoms, *Acta Chem. Scand.* 4 (1950) 1119–1130, <https://doi.org/10.3891/acta.chem.scand.04-1119>.
- [83] R. Enjalbert, J. Galy, A refinement of the structure of V₂O₅, *Acta Crystallogr. C* 42 (1986) 1467–1469, <https://doi.org/10.1107/S0108270186091825>.
- [84] V. Mokerov, V. Makarov, V. Tulvinskii, A. Begishev, Optical properties of vanadium pentoxide in the region of photon energies from 2 eV to 14 eV, *Optic Spectrosc.* 40 (1) (1976) 58–61.
- [85] N. Van Hieu, D. Lichtman, Bandgap radiation induced photodesorption from V₂O₅ powder and vanadium oxide surfaces, *J. Vac. Sci. Technol.* 18 (1) (1981) 49, <https://doi.org/10.1116/1.570698>.
- [86] S.F. Cogan, N.M. Nguyen, S.J. Perrotti, R.D. Rau, Optical properties of electrochromic vanadium pentoxide, *J. Appl. Phys.* 66 (3) (1989) 1333–1337, <https://doi.org/10.1063/1.344432>.
- [87] E.E. Chain, Optical properties of vanadium dioxide and vanadium pentoxide thin films, *Appl. Optic.* 30 (19) (1991) 2782–2787, <https://doi.org/10.1364/ao.30.002782>.
- [88] A.Z. Moshfegh, A. Ignatiev, Formation and characterization of thin film vanadium oxides: auger electron spectroscopy, X-ray photoelectron spectroscopy, X-ray diffraction, scanning electron microscopy, and optical reflectance studies, *Thin Solid Films* 198 (1–2) (1991) 251–268, [https://doi.org/10.1016/0040-6090\(91\)90344-W](https://doi.org/10.1016/0040-6090(91)90344-W).
- [89] Z.S. El Mandouh, M.S. Selim, Physical properties of vanadium pentoxide sol-gel films, *Thin Solid Films* 371 (1–2) (2000) 259–263, [https://doi.org/10.1016/S0040-6090\(00\)01003-8](https://doi.org/10.1016/S0040-6090(00)01003-8).

- [90] C.V. Ramana, B.S. Naidu, O.M. Hussain, R. Pinto, Low-temperature growth of vanadium pentoxide thin films produced by pulsed laser ablation, *J. Phys. D Appl. Phys.* 34 (7) (2001) L35–L38, <https://doi.org/10.1088/0022-3727/34/7/101>.
- [91] L.R. De Jesus, J.L. Andrews, A. Parija, S. Banerjee, Defining diffusion pathways in intercalation cathode materials: some lessons from V_2O_5 on directing cation traffic, *ACS Energy Lett.* 3 (4) (2018) 915–931, <https://doi.org/10.1021/acsenergylett.8b00156>.
- [92] V.V. Kulish, D. Koch, S. Manzhos, *Ab initio* study of Li, Mg and Al insertion into rutile VO_2 : fast diffusion and enhanced voltages for multivalent batteries, *Phys. Chem. Chem. Phys.* 19 (33) (2017) 22538–22545, <https://doi.org/10.1039/c7cp04360k>.
- [93] V.V. Kulish, O.I. Malyi, C. Persson, P. Wu, Phosphorene as an anode material for Na-ion batteries: a first-principles study, *Phys. Chem. Chem. Phys.* 17 (21) (2015) 13921–13928, <https://doi.org/10.1039/c5cp01502b>.
- [94] M.A. Sk, S. Manzhos, Exploring the sodium storage mechanism in disodium terephthalate as anode for organic battery using density-functional theory calculations, *J. Power Sources* 324 (2016) 572–581, <https://doi.org/10.1016/j.jpowsour.2016.05.101>.
- [95] J.M. Velazquez, S. Banerjee, Catalytic growth of single-crystalline V_2O_5 nanowire arrays, *Small* 5 (9) (2009) 1025–1029, <https://doi.org/10.1002/sml.200801278>.
- [96] Y. Cai, G. Fang, J. Zhou, S. Liu, Z. Luo, A. Pan, G. Cao, S. Liang, Metal-organic framework-derived porous shuttle-like vanadium oxides for sodium-ion battery application, *Nano Res* 11 (1) (2018) 449–463, <https://doi.org/10.1007/s12274-017-1653-9>.
- [97] Y. Wang, K. Takahashi, K. Lee, G. Cao, Nanostructured vanadium oxide electrodes for enhanced lithium-ion intercalation, *Adv. Funct. Mater.* 16 (9) (2006) 1133–1144, <https://doi.org/10.1002/adfm.200500662>.
- [98] J.M. Velazquez, C. Jaye, D.A. Fischer, S. Banerjee, Near edge X-ray absorption fine structure spectroscopy studies of single-crystalline V_2O_5 nanowire arrays, *J. Phys. Chem. C* 113 (18) (2009) 7639–7645, <https://doi.org/10.1021/jp900809y>.
- [99] C. Wu, Y. Xie, Promising vanadium oxide and hydroxide nanostructures: from energy storage to energy saving, *Energy Environ. Sci.* 3 (9) (2010) 1191–1206, <https://doi.org/10.1039/c0ee00026d>.
- [100] P.M. Marley, T.A. Abtew, K.E. Farley, G.A. Horrocks, R.V. Dennis, P. Zhang, S. Banerjee, Emptying and filling a tunnel bronze, *Chem. Sci.* 6 (3) (2015) 1712–1718, <https://doi.org/10.1039/c4sc03748k>.
- [101] J. Galy, D. Lavaud, A. Casalot, P. Hagenmuller, Les bronzes oxygénés de vanadium de formule $Cu_xV_2O_5$: I. Structure cristalline des phases $Cu_xV_2O_5\beta$ et $Cu_xV_2O_5\epsilon$, *J. Solid State Chem.* 2 (4) (1970) 531–543, [https://doi.org/10.1016/0022-4596\(70\)90048-4](https://doi.org/10.1016/0022-4596(70)90048-4).
- [102] J.-M. Savariault, E. Deramond, J. Galy, Vanadium oxide bronzes – the double mixed valence phases $Cu_{xT}+Cu_{x(1-\theta)}2+V_{(2-x)(2-\theta)}5+V_{x(2-\theta)}4+O_5$ A joint XRD and ESCA investigation, *Z. Kristallogr.* 209 (5) (1994) 405–412, <https://doi.org/10.1524/zkri.1994.209.5.405>.
- [103] C.J. Patridge, C. Jaye, H. Zhang, A.C. Marschilok, D.A. Fischer, E.S. Takeuchi, S. Banerjee, Synthesis, structural characterization, and electronic structure of single-crystalline $Cu_xV_2O_5$ nanowires, *Inorg. Chem.* 48 (7) (2009) 3145–3152, <https://doi.org/10.1021/ic802408c>.
- [104] J.L. Andrews, A. Mukherjee, H.D. Yoo, A. Parija, P.M. Marley, S. Fakra, D. Prendergast, J. Cabana, R.F. Klie, S. Banerjee, Reversible Mg-ion insertion in a metastable one-dimensional polymorph of V_2O_5 , *Chem* 4 (3) (2018) 564–585, <https://doi.org/10.1016/j.chempr.2017.12.018>.
- [105] J. Galy, J. Darriet, A. Casalot, J.B. Goodenough, Structure of the $M_xV_2O_5\beta$ and $M_xV_{2-\gamma}O_5\beta$ phases, *J. Solid State Chem.* 1 (3–4) (1970) 339–348, [https://doi.org/10.1016/0022-4596\(70\)90114-3](https://doi.org/10.1016/0022-4596(70)90114-3).
- [106] S. Liang, J. Zhou, G. Fang, C. Zhang, J. Wu, Y. Tang, A. Pan, Synthesis of mesoporous $\beta\text{-Na}_{0.33}V_2O_5$ with enhanced electrochemical performance for lithium ion batteries, *Electrochim. Acta* 130 (2014) 119–126, <https://doi.org/10.1016/j.electacta.2014.02.131>.
- [107] J.K. Kim, B. Senthilkumar, S.H. Sahngong, J.H. Kim, M. Chi, Y. Kim, New chemical route for the synthesis of $\beta\text{-Na}_{0.33}V_2O_5$ and its fully reversible Li intercalation, *ACS Appl. Mater. Interfaces* 7 (12) (2015) 7025–7032, <https://doi.org/10.1021/acsami.5b01260>.